CLEANER INDUSTRIES
GREENER TOMORROW

DATA ANALYTICS PROJECT REPORT

TRACING THE CARBON IMPACT

An Exploratory Analysis of Industry Emissions

Using data-driven insights to understand, compare and reduce the carbon footprint across industries.

DATASET
SupplyChainGHGEmissionFactors (v1.3.0)

TOOLS
Python | Pandas | Matplotlib
Seaborn | Jupyter Notebook

FOCUS
Industry-wise Emission Factors & Carbon Impact Analysis



Prepared by
Blessy

LESS EMISSIONS

ABSTRACT

Tracing the Carbon Impact is a Python-based exploratory data analytics project examining how greenhouse-gas emission intensity varies across U.S. industry classifications. The analysis uses the supplied Supply Chain GHG Emission Factors (v1.3.0) dataset, which contains 1,016 industry-level observations and eight original attributes. Emission factors are expressed in kilograms of carbon-dioxide equivalent per 2022 U.S. dollar of expenditure, enabling comparisons on a consistent monetary basis. The study investigates factors reported with and without margins, examines the distribution of values, identifies high- and low-intensity industry categories, and visualizes relationships and outliers. Data preparation and analysis are performed with Python and Pandas, while Matplotlib and Seaborn support statistical graphics. The results demonstrate substantial variation across industry categories and show how margin-inclusive and margin-exclusive measurements can differ. These values represent emissions per expenditure dollar—not the total emissions of an industry—and should be interpreted accordingly. The project provides an accessible, reproducible foundation for expenditure-based carbon-impact assessment and more informed supply-chain analysis.

Project at a Glance

INDICATOR	VALUE
Dataset records	1,016
Original attributes	8
Distinct NAICS industry titles	1016
Primary unit	kg CO_2e / 2022 USD
Analysis environment	Python / Jupyter Notebook
Notebook visualizations	12

Introduction and Problem Context

Supply-chain greenhouse-gas factors estimate emissions associated with monetary expenditure on products or services. Comparing factors across industries can help identify which purchasing categories have relatively high climate impacts per dollar. This report explores industry-level patterns using the provided NAICS-aligned dataset and the accompanying Python notebook. The emphasis is on transparent descriptive analytics and readable visual communication.

Problem Statement

Emission factors vary widely between industries. A systematic exploratory analysis is needed to understand their distribution, rank categories by factor magnitude, and assess the contribution of margins to purchaser-price emission factors.

Objectives and Scope

• Understand the structure and quality of the source data.

• Compare industries using emission factors with margins.

• Identify industries at the upper and lower ends of the factor distribution.

• Quantify the difference between factors with and without margins.

• Communicate relationships, skewness and outliers through visualizations.

Scope: This is a cross-sectional exploratory analysis of the supplied U.S. industry-factor dataset. It does not forecast future emissions, calculate organization-specific carbon footprints or establish causal relationships.

Dataset and Data Dictionary

Source file: SupplyChainGHGEmissionFactors_v1.3.0_NAICS_CO2e_USD2022 (1).csv. The dataset has 1,016 rows and eight original fields. Each row represents an industry title in the supplied classification. All rows report “All GHGs” in purchaser-price 2022-dollar emission-factor units.

FIELD	DESCRIPTION
2017 NAICS Code	Industry classification code
2017 NAICS Title	Industry or commodity label
GHG	Greenhouse-gas coverage
Unit	Factor measurement unit
Factors without Margins	Base supply-chain factor
Margins of Factors	Reported margin component
Factors with Margins	Purchaser-price factor
Reference USEEIO Code	Mapping to USEEIO sector

Tools and Analytical Method

Python is used to load and analyze the CSV with Pandas. Matplotlib and Seaborn support the charts, while Jupyter Notebook organizes the workflow. The notebook follows five broad stages: data understanding; cleaning and preprocessing; exploratory analysis; visualization; and insights and recommendations.

Method: inspect shape and data types; check missingness and duplicates; summarize numerical columns; derive margin measures; rank industries; compare distributions and relationships; and interpret findings against the stated unit.

Data Preparation and Quality Checks

The Python notebook inspects dataset dimensions, head and tail records, data types, null counts, duplicate counts and descriptive statistics. Two analytical fields are derived:

Margin Contribution = Emission Factor with Margins − Emission Factor without Margins

Margin Contribution (%) = 100 × Margin Contribution / Emission Factor with Margins

These calculated fields are used for comparative exploration. They are not part of the original eight-column CSV. The dataset's reported margin column is also retained as provided; small discrepancies between calculated differences and reported margins may arise from rounding.

DATA QUALITY CHECK	RESULT
Missing values across original fields	0
Exact duplicate original rows	0
Rows / columns	1,016 / 8
Industry titles (unique)	1016

Descriptive Statistics

MEASURE	WITHOUT MARGINS	WITH MARGINS
Mean	0.265	0.282
Median	0.159	0.173
Minimum	0.026	0.029
Maximum	3.846	3.924
Standard deviation	0.315	0.321

All values above are in kg CO_2e per 2022 USD. Summary statistics are calculated directly from the attached CSV.

Exploratory Analysis and Visualizations

Highest Emission-Factor Industries

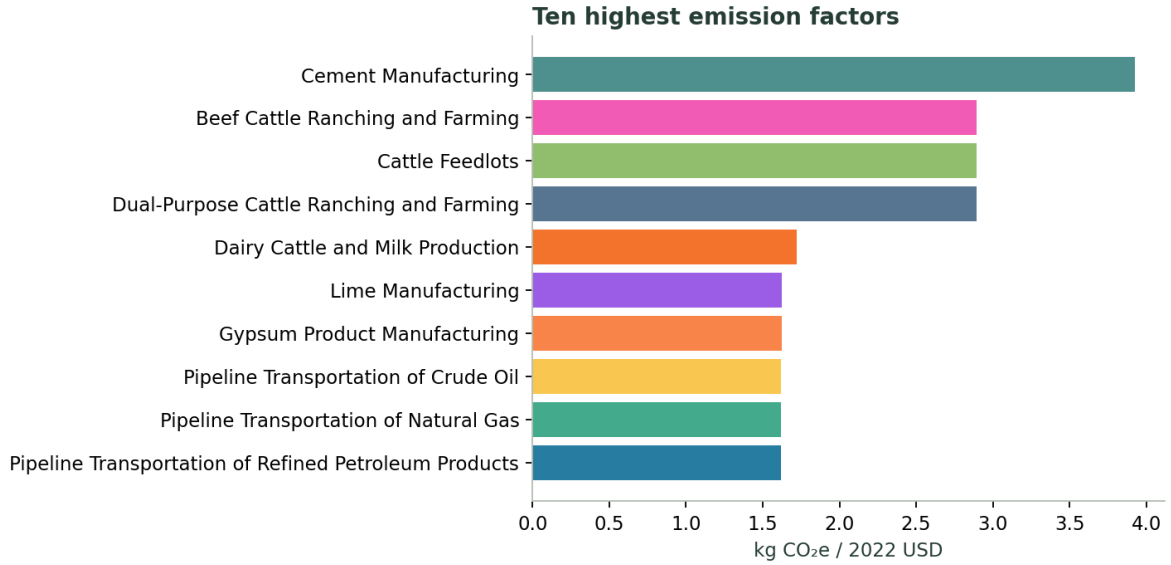


Figure 1. Ten highest supply-chain emission factors including margins.

The highest factor belongs to Cement Manufacturing, at 3.924 kg CO_2e per 2022 USD. This ranking reflects intensity per expenditure dollar, not total industry emissions.

Lowest Emission-Factor Industries

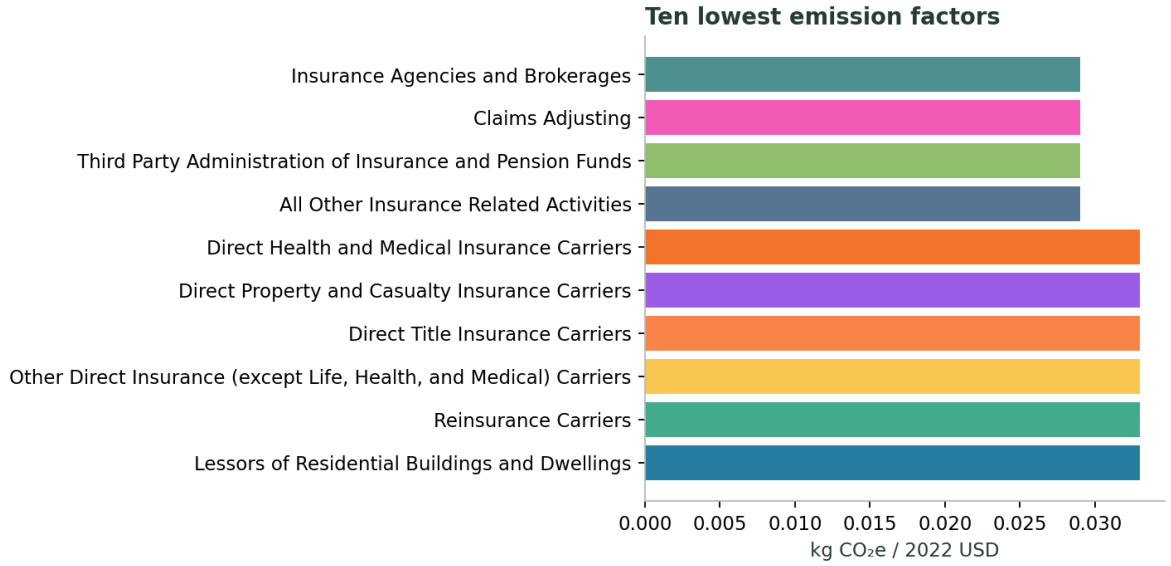


Figure 2. Ten lowest emission factors including margins.

The lower tail provides a contrast with high-factor categories and demonstrates the extent of industry-level variation.

Distribution of Carbon Impact

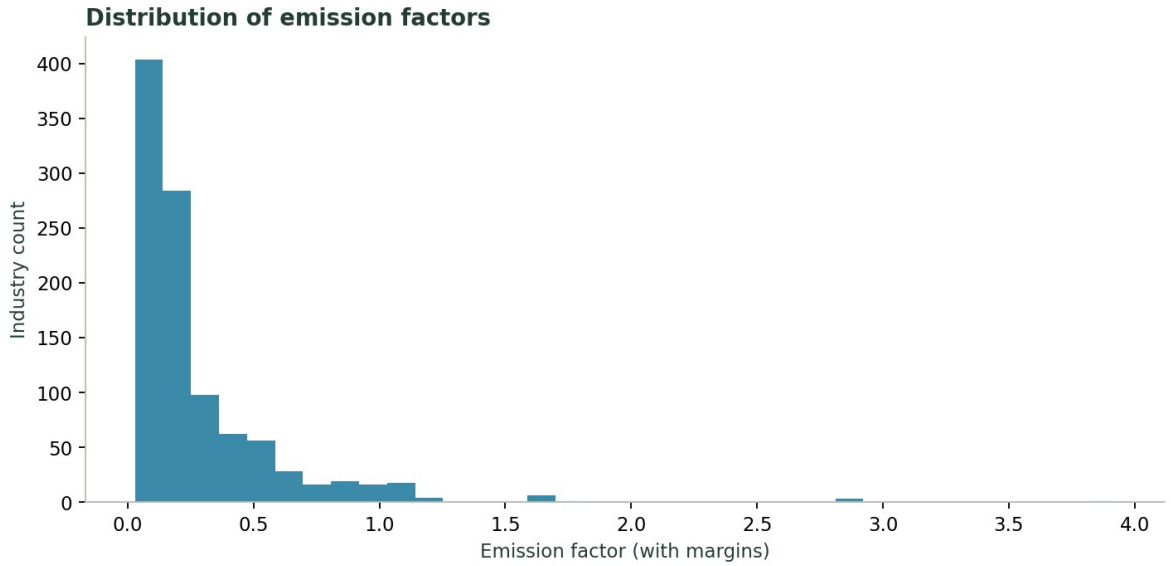


Figure 3. Histogram of the purchaser-price emission factor.

The mean factor (0.282) exceeds the median (0.173), consistent with a right-skewed distribution in which a smaller number of industries have substantially higher factors.

Relationship Between Base and Total Factors

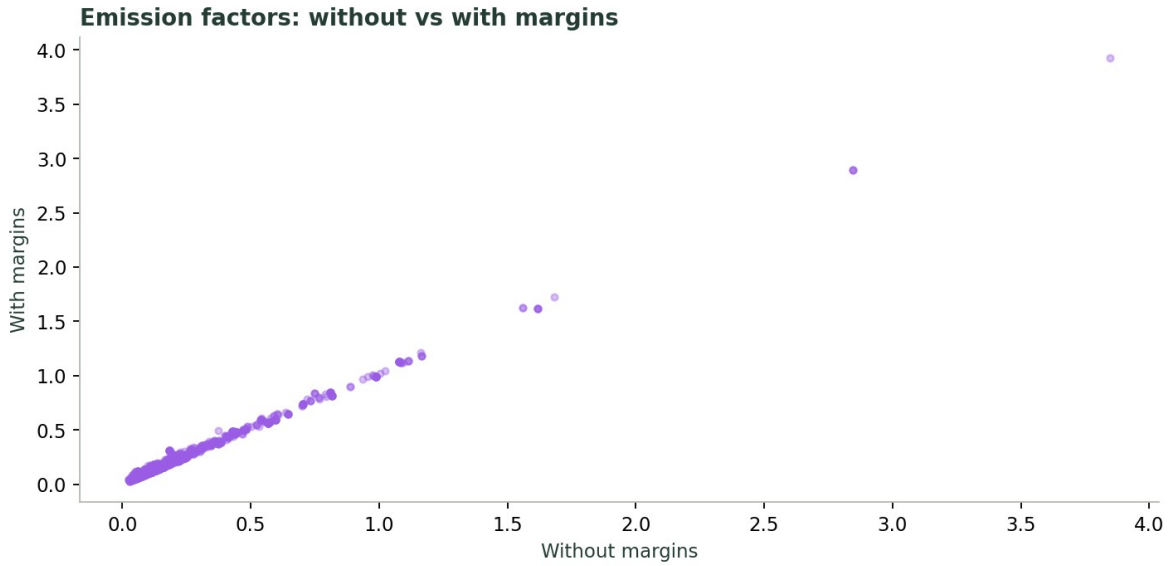


Figure 4. Relationship between factors excluding and including margins.

The two measures show a strong positive association. Most industry factors rise when margins are included, but the size of the increase differs between categories.

Outliers and Dispersion



Figure 5. Box plot showing spread and high-end outliers.

The box plot makes the concentration of observations at lower factor values and the presence of high-end outliers easier to see.

Correlation Structure

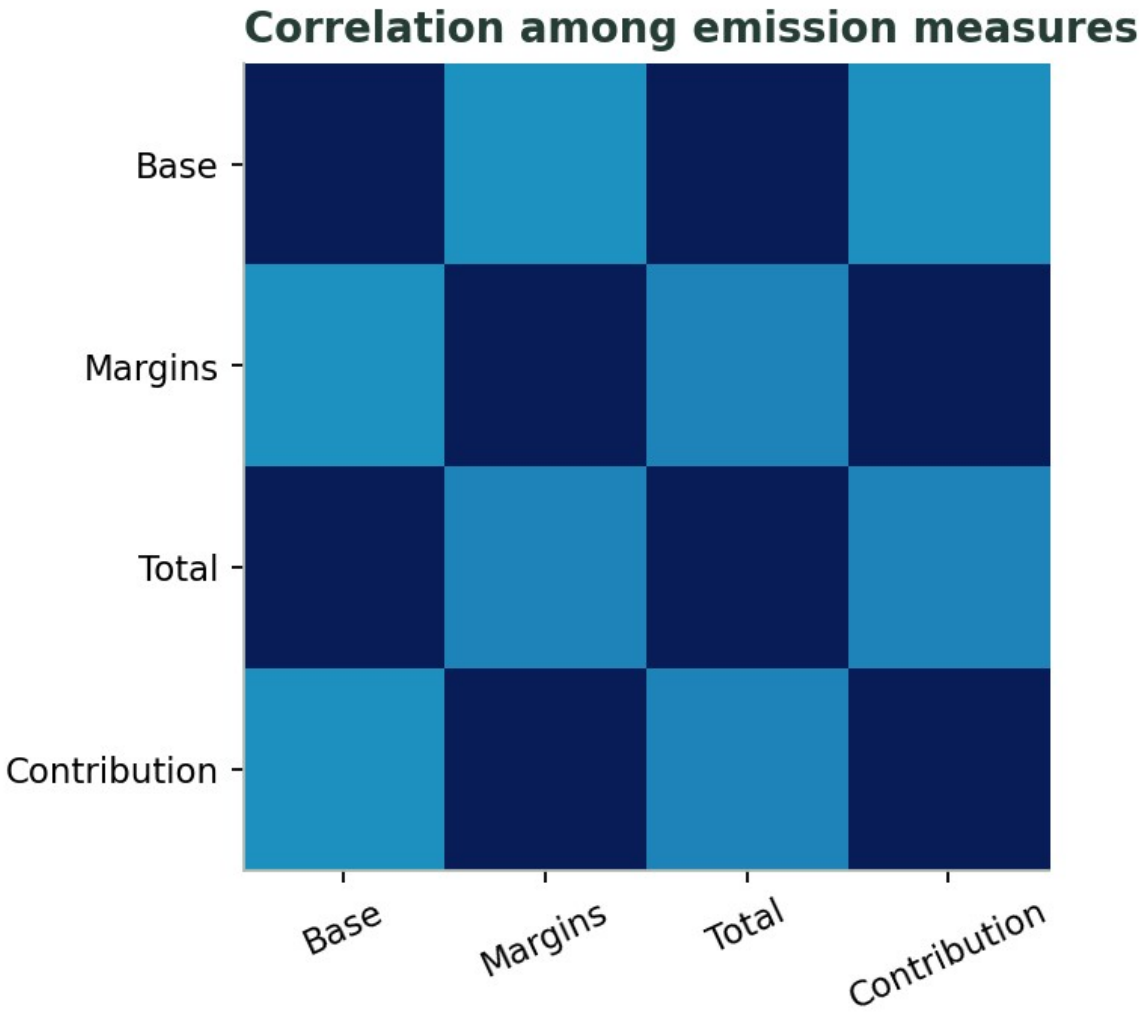


Figure 6. Correlation matrix of selected factor measures.

Correlation is descriptive and does not demonstrate causation. The calculated contribution and reported margin column represent closely related concepts but may not be identical because the source factors are rounded.

Highest Margin Contributions

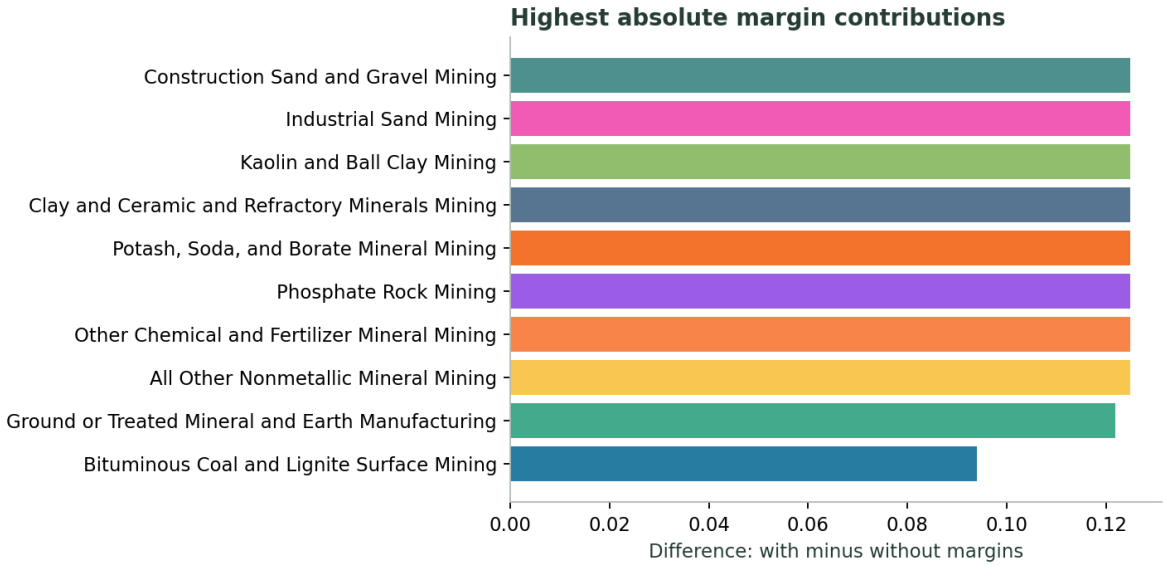


Figure 7. Ten largest calculated differences between factors with and without margins.

This comparison identifies where the difference between the two factor definitions is largest in absolute terms.

Margin Contribution as a Percentage

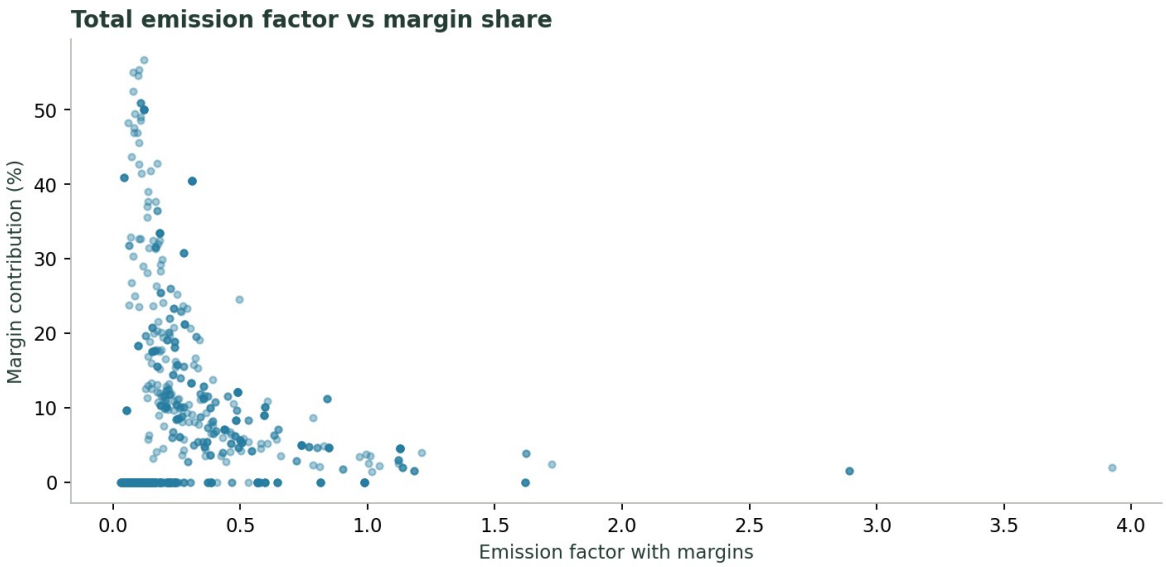


Figure 8. Total emission factor against calculated margin contribution percentage.

The percentage measure allows relative comparisons but should be interpreted alongside the absolute difference, particularly when the overall factor is small.

Composition of High-Factor Industries

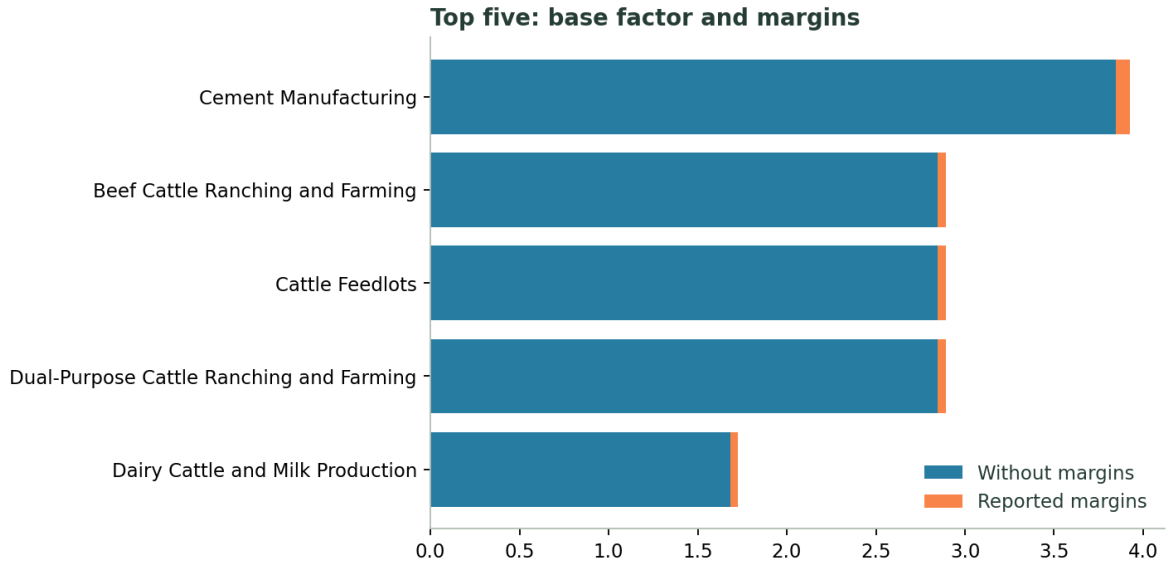


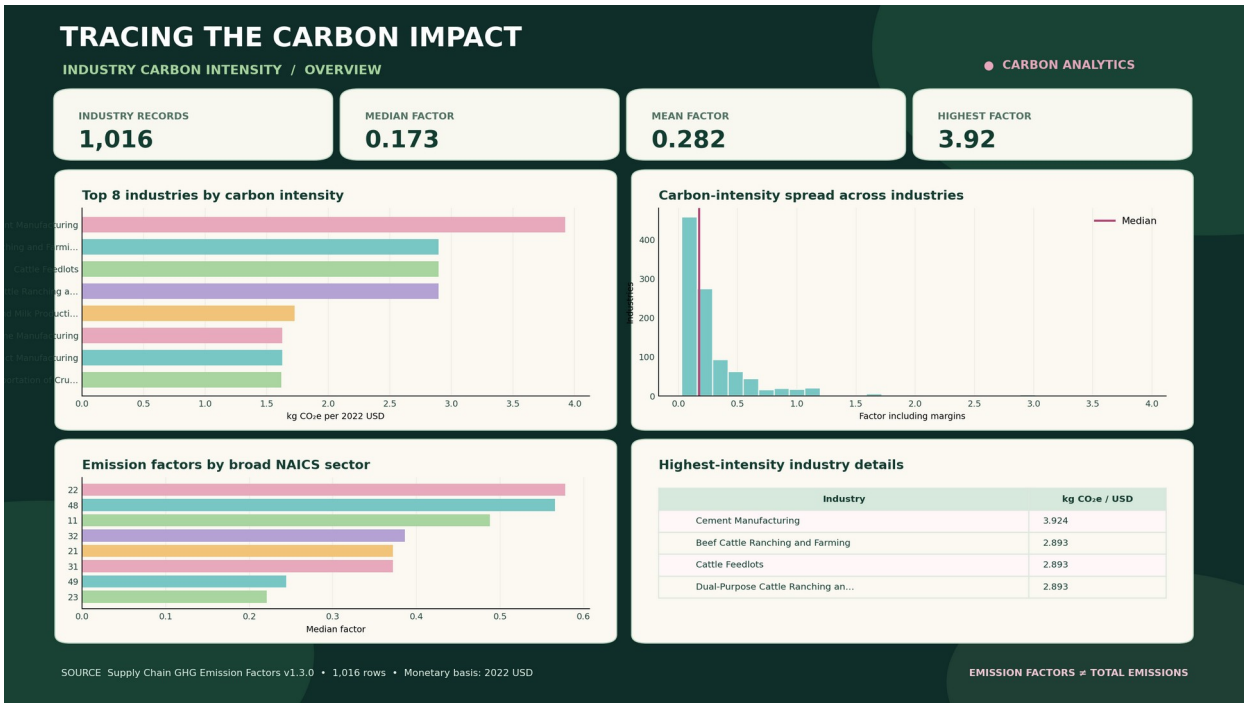
Figure 9. Reported base and margin components for the five highest-factor industries.

The component view distinguishes the base supply-chain factor from the separately reported margin factor for the leading industries.

Carbon Impact — Dashboard Gallery

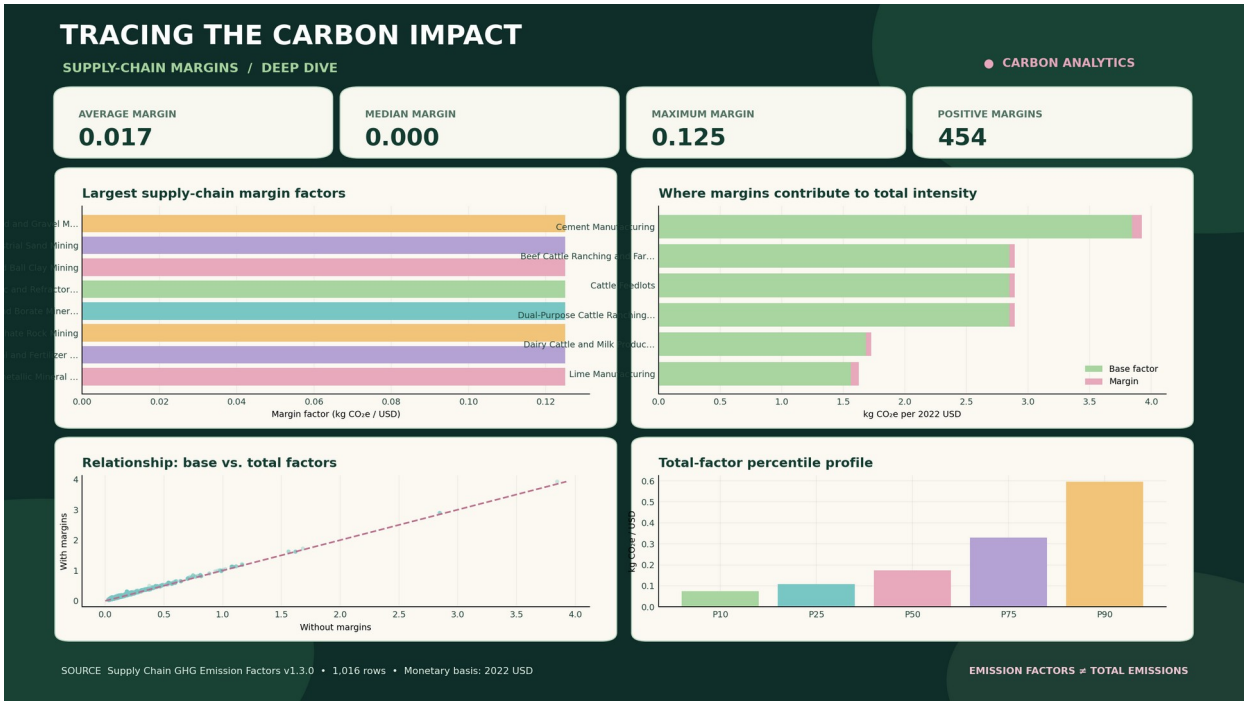
The following two dashboard views summarize distinct aspects of the supplied industry-level emission-factor data. Each dashboard is a static visual summary, not an interactive Power BI report.

Industry Carbon Intensity Overview



This view compares the highest-intensity industries, overall distribution, sector-level medians, and industry details.

Supply-Chain Margin Analysis



This view isolates margin contributions, compares base and total factors, and shows the percentile profile.

Findings and Interpretation

• The dataset covers 1,016 unique industry titles and uses a common expenditure-based unit.

• The highest recorded purchaser-price factor is 3.924 kg CO_2e / 2022 USD (Cement Manufacturing).

• The median purchaser-price factor is 0.173, compared with a mean of 0.282.

• Emission-factor rankings reveal relative carbon intensity, not total emissions or total sector size.

• The margin analysis shows why factor definitions should be kept consistent when comparing purchasing categories.

Limitations and Recommendations

Limitations: the dataset contains emission factors rather than observed emissions, spending volumes or longitudinal measurements. Values depend on source-model assumptions, NAICS classifications and purchaser-price conventions. Differences in rounded columns should not be overinterpreted. The analysis is exploratory and cannot establish that any industry is environmentally better overall.

Recommendations: use a consistent factor definition; document the expenditure year and unit; pair factors with actual purchasing data before estimating footprints;

prioritize review of high-intensity procurement categories; and validate conclusions against updated source documentation before operational decisions.

Conclusion and Future Work

The project demonstrates a reproducible Python workflow for exploring industry-level greenhouse-gas emission factors. The supplied data show pronounced differences between categories and a positively skewed distribution of purchaser-price factors. Rankings, distribution plots, correlation analysis and margin comparisons together offer a clearer view of relative carbon intensity. Future work could integrate organization-specific expenditure, introduce interactive dashboards, compare multiple factor releases and add uncertainty analysis where supporting data are available.

References and Appendix

Dataset: Supply Chain GHG Emission Factors v1.3.0, NAICS / CO_2e / USD 2022 (provided CSV).

Project source: Tracing_The_Carbon_Impact.ipynb (provided Python/Jupyter notebook).

Software used in notebook: Python, Pandas, Matplotlib, Seaborn and Jupyter Notebook.

Appendix A — Notebook Visualization Inventory

- Top 10 industries
- Bottom 10 industries
- Distribution of carbon impact
- Without vs with margins
- Top-five carbon-impact share
- Box plot and outliers
- Correlation heatmap
- Composition of summed factors
- Highest margin contribution
- Distribution by measure
- Top-five factor components
- Carbon impact vs margin contribution %

Note: The report reproduces nine core visualizations from the project's analytical themes. The original notebook includes 12 chart sections, including aggregate

composition and top-five share views. This report uses the uploaded dataset to generate its figures.

Project Source Files

The original project materials are provided with this report for review, reproducibility and further analysis. Select the file names below to open the corresponding files from the accompanying project package.

PYTHON PROJECT NOTEBOOK

Open Python analysis notebook (.ipynb)

Includes the Python-based exploratory analysis, data processing and visualization workflow used for this project.

SOURCE DATASET

Open carbon-emissions dataset (.csv — Excel-compatible)

The dataset is provided in CSV format and can be opened directly in Microsoft Excel or imported into Python using Pandas.

Access note: For reliable access, extract the complete project package into one folder before opening the report. The PDF also includes both source files as embedded attachments, accessible through the PDF reader's attachments panel.